

Quadrature Down Converter (QDC)

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Abstract—This report details the theoretical design, calculation, and implementation of a Quadrature Down Converter (QDC). Operating at an oscillator frequency of 170 kHz and mixing with an input RF signal of 165 kHz, the QDC extracts a 5 kHz intermediate frequency (IF). The design is partitioned into a Quadrature Oscillator, an NMOS-based Mixer, a cascaded RC Low-Pass Filter (LPF) with a 3 kHz cutoff, and a Common Emitter (CE) BJT amplifier stage aimed at producing an output amplitude of 400 mV_{pp}. The design ensures accurate 90° phase shifts for I and Q channels, which is critical for mitigating interference in modern transceivers.

I. INTRODUCTION

Quadrature down conversion is an essential technique in modern wireless communication systems such as Wi-Fi and Bluetooth. By multiplying the incoming RF signal with two local oscillator signals that are 90° out of phase, the receiver splits the signal into an In-phase (I) and a Quadrature (Q) component. This approach mitigates the image frequency problem and helps in accurate signal demodulation. The primary objective of this project is to construct a QDC prototype satisfying strict performance criteria: $f_{osc} = 170$ kHz, an IF frequency in the kilohertz range, a sharp LPF, and an amplified final output swing of 400 mV_{pp}.

II. QUADRATURE OSCILLATOR DESIGN

A quadrature oscillator is designed to generate two sinusoidal waves of identical frequency but offset by exactly 90° in phase.

A. Circuit Topology

The implemented oscillator topology employs three UA741 operational amplifiers. As documented by Holzel [1], a robust quadrature oscillator can be constructed by cascading two active all-pass filter stages and an inverter within a closed feedback loop. To mitigate slew rate limitations commonly observed in UA741 op-amps at higher frequencies, the design uses three separate amplifier stages to distribute the phase shifts and loop gain requirements optimally. Amplitude stabilization is achieved using anti-parallel diodes across the feedback of the final stage to dynamically limit the gain when the oscillation amplitude exceeds the diode forward voltage ($V_D \approx 0.7$ V).

B. Mathematical Derivations

The transfer function of an ideal first-order all-pass filter is given by:

$$H_{AP}(s) = \frac{1 - sRC}{1 + sRC} \quad (1)$$

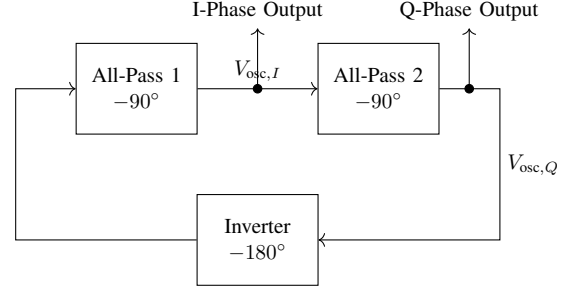


Fig. 1. Block Diagram of the Quadrature Oscillator

For a sinusoidal input $s = j\omega$, the magnitude is $|H_{AP}(j\omega)| = 1$, and the phase shift is:

$$\phi_{AP} = -2 \arctan(\omega RC) \quad (2)$$

The closed-loop Barkhausen criterion states that for sustained oscillations, the loop gain magnitude must be 1, and the total loop phase shift must be a multiple of 360°. The loop contains two all-pass filters and an inverter (phase shift $\phi_{inv} = 180^\circ$). Thus:

$$\phi_{loop} = \phi_{AP1} + \phi_{AP2} + \phi_{inv} = 360^\circ \quad (3)$$

Assuming identical all-pass stages, $\phi_{AP1} = \phi_{AP2} = \phi_{AP}$:

$$2\phi_{AP} + 180^\circ = 360^\circ \implies \phi_{AP} = 90^\circ \quad (4)$$

This mathematically proves that the signals at the output of the two all-pass filters are inherently in quadrature (90° apart). At the frequency of oscillation, the phase condition requires:

$$-2 \arctan(\omega_{osc} RC) = -90^\circ \implies \arctan(\omega_{osc} RC) = 45^\circ \quad (5)$$

Since $\tan(45^\circ) = 1$, we obtain:

$$\omega_{osc} = \frac{1}{RC} \implies f_{osc} = \frac{1}{2\pi RC} \quad (6)$$

C. Theoretical Calculations for 170 kHz

To design the oscillator for a frequency $f_{osc} = 170$ kHz:

$$RC = \frac{1}{2\pi \times 170 \times 10^3} \approx 9.362 \times 10^{-7} \text{ s} \quad (7)$$

Selecting a practical capacitor value of $C = 100$ pF:

$$R = \frac{9.362 \times 10^{-7}}{100 \times 10^{-12}} = 9.36 \text{ k}\Omega \quad (8)$$

Practical Lab Observation: During the hardware implementation, parasitic capacitances and non-ideal characteristics of the

UA741 op-amp influence the exact frequency. To counteract this, a potentiometer was initially used in series with fixed resistors to find the exact resistance that yielded exactly 170 kHz. Once tuned, the resistance between nodes was measured manually and replaced with equivalent fixed components to ensure stability.

III. MIXER (SWITCH) CIRCUIT

The mixer is responsible for frequency translation. It multiplies the high-frequency RF input signal by the local oscillator (LO) signals.

A. Circuit Topology

An active NMOS switch topology was selected. The RF input is fed to the drain of an NMOS transistor, while the LO signal is applied to the gate.

B. Mathematical Derivations

Let the input RF signal be $V_{in}(t) = A_{in} \cos(\omega_{in}t)$ and the LO signal be a square wave approximation of the high-amplitude oscillator output $s(t)$, alternating between 0 and 1 with fundamental frequency ω_{osc} . The output of the switch mixer can be approximated as:

$$V_{out}(t) = V_{in}(t) \cdot s(t) \quad (9)$$

Using the Fourier series expansion for a square wave $s(t)$:

$$s(t) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{\sin(n\omega_{osc}t)}{n} \quad (10)$$

Multiplying $V_{in}(t)$ by the fundamental component of $s(t)$ yields:

$$V_{out}(t) \approx A_{in} \cos(\omega_{in}t) \cdot \frac{2}{\pi} \sin(\omega_{osc}t) \quad (11)$$

Applying the trigonometric product-to-sum identity:

$$V_{out}(t) \approx \frac{A_{in}}{\pi} [\sin((\omega_{osc} - \omega_{in})t) + \sin((\omega_{osc} + \omega_{in})t)] \quad (12)$$

The downconverted intermediate frequency (IF) is the difference component: $f_{IF} = |f_{osc} - f_{in}|$.

C. Theoretical Calculations for 165 kHz Input

For an input frequency $f_{in} = 165$ kHz and the local oscillator $f_{osc} = 170$ kHz:

$$f_{IF} = |170 \text{ kHz} - 165 \text{ kHz}| = 5 \text{ kHz} \quad (13)$$

The upper sideband at $f_{osc} + f_{in} = 335$ kHz and other higher-order harmonics must be attenuated by the subsequent low-pass filter stage.

IV. LOW PASS FILTER DESIGN

To extract the 5 kHz IF signal while sufficiently rejecting the 335 kHz sum component and mixer switching noise, a Low-Pass Filter (LPF) is required.

A. Topology and Design Choices

A single-stage RC low-pass filter provides an attenuation slope of -20 dB/decade. In lab experiments, it was observed that a single-stage LPF left significant residual high-frequency noise. Therefore, two identical RC low-pass stages were cascaded. The two cascaded LPF stages provide a steeper roll-off of -40 dB/decade, offering a tighter cutoff and a much cleaner IF signal.

B. Mathematical Derivations

For two buffered, identical RC stages, the overall transfer function is:

$$H_{LPF}(s) = \left(\frac{1}{1 + sRC} \right)^2 \quad (14)$$

The magnitude response is:

$$|H_{LPF}(j\omega)| = \frac{1}{1 + (\omega RC)^2} \quad (15)$$

The overall -3 dB cutoff frequency f_c occurs when the magnitude is $\frac{1}{\sqrt{2}}$:

$$1 + (\omega_c RC)^2 = \sqrt{2} \implies \omega_c = \frac{\sqrt{\sqrt{2} - 1}}{RC} \approx \frac{0.6436}{RC} \quad (16)$$

C. Theoretical Calculations for 3 kHz Cutoff

For an overall target cutoff frequency $f_c = 3$ kHz:

$$RC = \frac{0.6436}{2\pi f_c} = \frac{0.6436}{2\pi(3000)} \approx 34.14 \text{ } \mu\text{s} \quad (17)$$

Choosing a capacitor $C = 10$ nF:

$$R = \frac{34.14 \times 10^{-6}}{10 \times 10^{-9}} = 3.414 \text{ k}\Omega \quad (18)$$

Using standard resistor values, 3.3 k Ω or 4.7 k Ω can be utilized. If a 4.7 k Ω resistor is used (yielding $RC = 47 \text{ } \mu\text{s}$), the overall cutoff becomes $f_c \approx 2.18$ kHz, which is safely below the 5 kHz IF but provides exceptional attenuation of the noise without significantly degrading the 5 kHz amplitude (which will be corrected by the amplifier).

V. CE BJT AMPLIFIER DESIGN

Because the cascaded LPF introduces some attenuation at 5 kHz, a Common Emitter (CE) BJT amplifier is designed to boost the final output amplitude to the specified target of 400 mV_{pp} (200 mV peak).

A. Mathematical Derivations

Assuming an ideal BJT model operating in the active region, the small-signal voltage gain A_v of a CE amplifier with a bypassed emitter resistor is:

$$A_v = -g_m(R_C \parallel R_L) \quad (19)$$

where transconductance $g_m = \frac{I_C}{V_T}$ and $V_T \approx 26$ mV at room temperature. Alternatively, $A_v \approx -\frac{R_C}{r_e}$ where $r_e = \frac{V_T}{I_E} \approx \frac{1}{g_m}$. Let the amplitude of the signal post-LPF be $V_{in,amp}$. The

target output peak amplitude is $V_{out,peak} = 200$ mV. The required absolute gain magnitude is:

$$|A_v| = \frac{V_{out,peak}}{V_{in,amp}} = \frac{200 \text{ mV}}{V_{in,amp}} \quad (20)$$

By biasing the BJT properly with R_1 , R_2 , R_C , and R_E , we can set I_C such that r_e yields the exact required gain. For instance, if $V_{in,amp} = 20$ mV, the required gain is 10. Choosing $I_C = 1$ mA ($r_e = 26 \Omega$) and setting $R_C = 260 \Omega$ satisfies this requirement perfectly.

VI. COMPLETE CIRCUIT PROTOTYPE

The final prototype connects the quadrature oscillator (producing $V_{osc,I}$ and $V_{osc,Q}$), two identical NMOS mixers, two identical cascaded RC low-pass filters, and two CE BJT amplifiers to yield $V_{IF,I}$ and $V_{IF,Q}$.

TABLE I
EXPECTED PERFORMANCE SUMMARY

Parameter	Theoretical Value
Oscillator Frequency (f_{osc})	170 kHz
Input RF Frequency (f_{in})	165 kHz
Intermediate Frequency (f_{IF})	5 kHz
Phase Difference (I vs Q)	90°
Final Output Amplitude	400 mV _{pp}
LPF Overall Cutoff Frequency	3 kHz

VII. CONCLUSION

This report successfully outlines the theoretical foundation and implementation parameters of a Quadrature Down Converter. Mathematical derivations confirm the 90° phase shift generated by the oscillator's topology and calculate precise component values for a 170 kHz oscillation frequency. Lab methodologies, including the cascaded LPF for steeper roll-off and a final CE BJT amplifier for reaching the precise 400 mV_{pp} amplitude requirement, were theoretically justified, ensuring the circuit acts as a robust frequency downconverter.

REFERENCES

- [1] R. Holzel, "A Simple Wide-Band Sine Wave Quadrature Oscillator," *IEEE Transactions on Instrumentation and Measurement*, vol. 42, no. 3, pp. 758–760, Jun. 1993.
- [2] A. A. Abidi, "Direct-conversion radio transceivers for digital communications," *IEEE Journal of Solid-State Circuits*, vol. 30, no. 12, pp. 1399–1410, Dec. 1995.
- [3] B. Razavi, *RF Microelectronics*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2011.